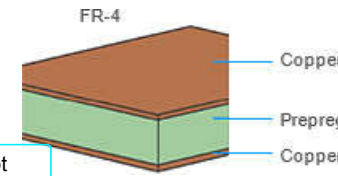


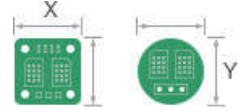
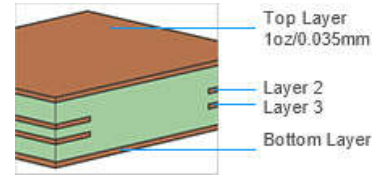
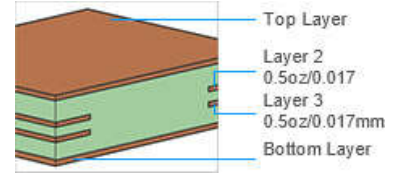
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PCB Specifications

Features	Capability	Notes	Patterns
Layer count	1,2,4,6 layers	The number of copper layers in the board.	
Controlled Impedance	4/6 layer, default layer stack-up	Controlled Impedance PCB Layer Stackup (https://cart.jlcpb.com/impedance) JLCPCB Impedance Calculator (https://cart.jlcpb.com/impedanceCalculation)	
Material	FR-4	FR-4 Standard Tg 130-140/ Tg 155	 FR-4 Copper Prepreg Copper

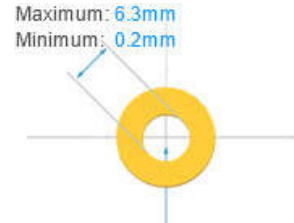
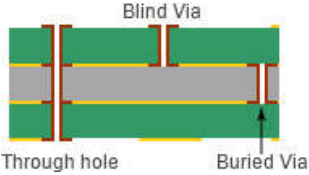
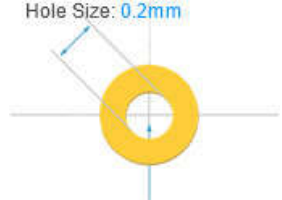
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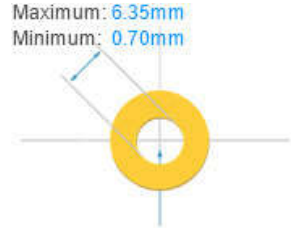
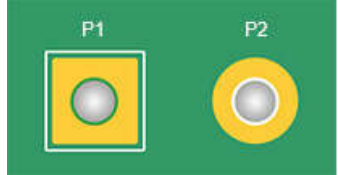
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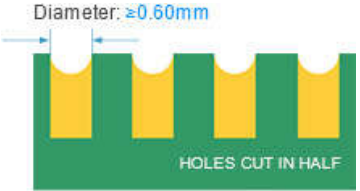
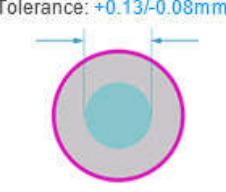
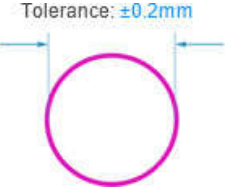
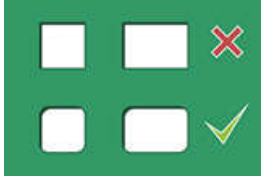
Features	Capability	Notes	Patterns
Dielectric constant	4.5(double-side PCB)	7628 structure 4.6 2313 structure 4.05 2116 structure 4.25	
Max. Dimension	400x500mm	The maximum dimension JLPCB can accept	
Dimension Tolerance	±0.2mm	±0.2mm for CNC routing, and ±0.4mm for V-scoring	
Board Thickness	0.4/0.6/0.8/1.0/1.2/1.6/2.0mm	The thickness of finished board.	
Thickness Tolerance (Thickness≥1.0mm)	± 10%	e.g. For the 1.6mm board thickness, the finished board thickness ranges from 1.44mm(T-1.6×10%) to 1.76mm(T+1.6×10%)	
Thickness Tolerance (Thickness<1.0mm)	± 0.1mm	e.g. For the 0.8mm board thickness, the finished board thickness ranges from 0.7mm(T-0.1) to 0.9mm(T+0.1).	
Finished Outer Layer Copper	1 oz/2 oz (35um/75um)	Finished copper weight of outer layer is 1oz or 2oz.	
Finished Inner Layer Copper	0.5 oz (17um)	Finished copper weight of inner layer is 0.5oz only.	

Drill/Hole Size

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Features	Capability	Notes	Patterns
Drill Hole Size (Mechanical)	0.20mm- 6.30mm	Min. drill size is 0.20mm. Max. drill size is 6.30mm.	
Drill Hole Size Tolerance	+0.13/-0.08mm	e.g. for the 0.6mm hole size, the finished hole size between 0.52mm to 0.73mm is acceptable.	
Blind/Buried Vias	Don't support	Currently we don't support Blind/Buried Vias, only make through holes.	
Min. Via hole size	0.2mm	For Single&Double Layer PCB, the minimum via hole size is 0.3mm;For Multi Layer PCB, the minimum via hole size is 0.2mm	
Min. Via diameter	0.45mm	For Single&Double Layer PCB, the minimum Via diameter is 0.6mm;For Multi Layer PCB, the minimum via diameter is 0.45mm.	

Features	Capability	Notes	Patterns
PTH hole Size	0.20mm - 6.35mm	The annular ring size will be enlarged to 0.15mm in production.	 Maximum: 6.35mm Minimum: 0.20mm
Pad Size	0.70mm- 6.35mm	The pad hole size will be enlarged 0.15mm in production.	 Maximum: 6.35mm Minimum: 0.70mm
Min. Non-plated holes	0.50mm	The minimum NPTH dimension is 0.50mm, Please add the NPTH in the mechanical layer or keep out layer.	 P1 P2
Min. Plated Slots	0.65mm	The minimum plated slot width is 0.65mm, which is drawn with a pad.	 Width ≥ 0.65mm
Min. Non-Plated Slots	1.0mm	The minimum Non-Plated Slot Width is 1.0mm, please draw the slot outline in the mechanical layer(GML or GKO)	 Width ≥ 1.0mm

Features	Capability	Notes	Patterns
Min. Castellated Holes	0.60mm	The minimum diameter of castellated holes is 0.60mm.	
Hole size Tolerance (Plated)	+0.13mm/-0.08mm	e.g. for the 1.00mm Plated hole, the finished hole size between 0.92mm to 1.13mm is acceptable.	
Hole size Tolerance (Non-Plated)	±0.2mm	e.g. for the 1.00mm Non-Plated hole, the finished hole size between 0.80mm to 1.20mm is acceptable.	
Rectangle Hole/Slot	Don't support	We don't make rectangle hole/slot, the rectangle hole/slot will be made as round or oval hole/slot by default.	

Minimum Annular Ring

	Minimum annular ring	PTH	Patterns
1oz Copper	0.13mm	0.3mm	We use cookies to offer you a better experience. Detailed information on the use of cookies on this website is provided in our Privacy Policy (https://support.jlcpb.com/article/59-privacy-policy). By using this site, you consent to the use of our cookies. Accept

	Minimum annular ring	PTH	Patterns
2oz Copper	0.2mm	0.3mm	Min. Annular Ring: 0.13mm

Minimum clearance

-

Features	Capabilities	Patterns
Hole to hole clearance(Different nets)	0.54mm	
Via to Via clearance(Same nets)	0.254mm	
Pad to Pad clearance(Pad without hole, Different nets)	0.127mm	
Pad to Pad clearance(Pad with hole, Different nets)	0.54mm	

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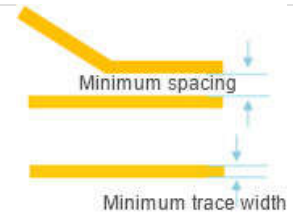
Features	Capabilities	Patterns
Via to Track	0.254mm	
PTH to Track	0.33mm	
NPTH to Track	0.254mm	
Pad to Track	0.2mm	

Minimum trace width and spacing

Copper weight	Min. Trace width	Min. Spacing	Patterns
H/HOZ (Inner layer)	5mil (0.127mm)	5mil (0.127mm)	
1oz (Outer layer)	1/2 layers: 5mil (0.127mm) 4/6 layers: 3.5mil (0.09mm)	1/2 layers: 5mil (0.127mm) 4/6 layers: 3.5mil (0.09mm)	

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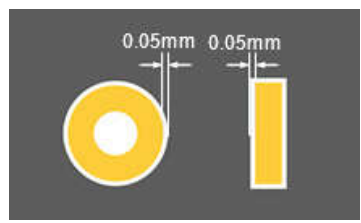
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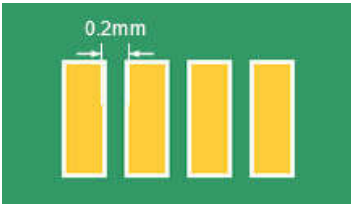
Copper weight	Min. Trace width	Min. Spacing	Patterns
2oz (Outer layer)	8mil (0.2mm)	8mil (0.2mm)	 The diagram illustrates the minimum spacing and minimum trace width for copper patterns. It shows two horizontal yellow lines representing traces. The distance between the two traces is labeled 'Minimum spacing'. The width of one of the traces is labeled 'Minimum trace width'.

BGA

Layer count	Min. BGA Pad Dimensions	Min. Distance Between BGA	Patterns
1/2 layers	0.4 mm	0.127mm	 The diagram shows a grid of yellow circular BGA pads. The distance between the centers of two adjacent pads is labeled '0.127mm'. The diameter of one pad is labeled '0.25mm'.
4/6 layers	0.25 mm	0.127mm	

Solder Mask

Features	Capabilities	Notes	Patterns
Solder mask opening/ expansion	0.05mm	The solder mask should have a minimum of a 0.05 mm "growth/mask opening" around the pad to allow for any mis-registration.	 The diagram shows a yellow circular pad and a yellow rectangular pad. The distance between the edge of the pad and the edge of the solder mask opening is labeled '0.05mm' for both shapes.

Features	Capabilities	Notes	Patterns
Solder bridge	0.2mm(green) 0.254mm(other colors)	To have solder mask bridge, the spacing between copper pads edge must be 0.2mm (8mils) or more.	
Solder mask color	green, red, yellow, blue, white, and black.	We use LPI (Liquid Photo Imageable) solder mask. It is the most common type of mask used today.	
Solder mask dielectric constant	3.8		
Solder mask thickness	10-15UM		

Legend

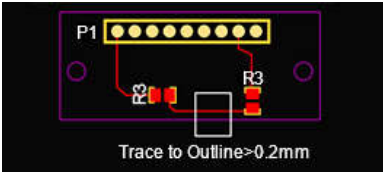
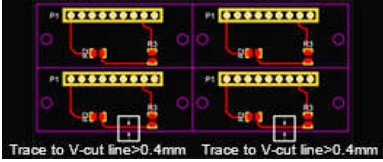
Features	Capabilities	Notes	Patterns
Minimum Line Width	6 mil (0.153mm)	Characters width less than 6mil(0.153mm) will be unidentifiable.	
Minimum text height	32 mil (0.8mm)	Characters height less than 32mil(0.8mm) will be unidentifiable.	

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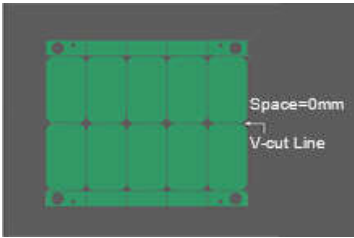
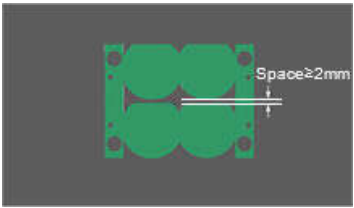
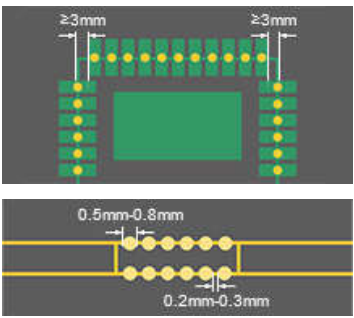
Features	Capabilities	Notes	Patterns
Character width to height ratio	1:6	The preferred ratio of width to height is 1:6.	
Pad To Silkscreen	0.15mm	The Minimum Distance Between Pad and Silkscreen is 0.15mm.	

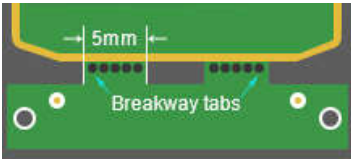
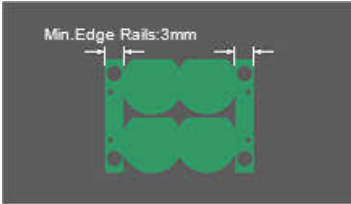
Board Outlines

Features	Capabilities	Notes	Patterns
Trace to Outline	0.2mm	Ships as individual board(Rounting):Trace to Outline≥0.2mm	
Trace to V-cut line	0.4mm	Ship as panel with V-scoring: Trace to V-cut line≥0.4mm	

Panelization

Features	Capabilities	Notes	Patterns
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Features	Capabilities	Notes	Patterns
Panelization without space	0mm	The space between boards is 0mm.	
Panelization with space	2mm	Make sure the space between boards should be $\geq 2\text{mm}$, otherwise it will be hard to process for rounding.	
Panelized Round board	$\geq 20\text{mm} \times 20\text{mm}$	The single round board size should be $\geq 20\text{mm} \times 20\text{mm}$. Panelize with stamp holes and add tooling strips on four board edges	
Panelized castellated holes board	Panelize with stamp holes and add tooling strips on four board edges	The distance between castellated hole and board corner should be larger than 3mm. Recommended diameter of stamp hole is 0.5mm-0.8mm; Recommended distance between the two stamp holes is 0.2-0.3mm	

Features	Capabilities	Notes	Patterns
Min. Width of Breakaway Tab	3mm	The minimum width of breakaway tab is 3mm. For breakaway with mouse-bites, the minimum width is 5mm.	
Min. Edge Rails	3mm	If choosing panel by JLCPCB, we will add 5mm edge rails on both sides by default.	

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